Heat Flow, Geometry, and Ricci Flow

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Overview of Presentation

- Heat flow acts as a smoothing process.
- The behavior of heat depends on the underlying space.
- Geometry is encoded by a metric.
- Curvature measures how that geometry "bends".
- Ricci flow: evolve the metric the same way heat evolves temperature.

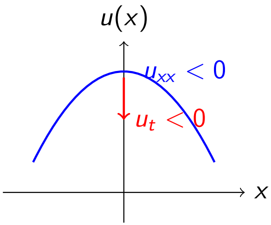
1D Heat Flow and the Laplacian

Before looking at surfaces, consider the 1D heat equation:

$$\frac{\partial u}{\partial t} = u_{xx}.$$

Why the second derivative? It measures *concavity*.

- If $u_{xx} < 0$ (concave down), temperature is a local maximum $\Rightarrow$ heat flows away ($u_t < 0$).
- If $u_{xx} > 0$ (concave up), temperature is a local minimum $\Rightarrow$ heat flows in ($u_t > 0$).



Heat Equation in $\mathbb{R}^2$

We expand this to the classical heat equation in Cartesian coordinates:

$$\frac{\partial u}{\partial t} = u_{xx} + u_{yy} = \Delta u.$$

- $u(x, y, t) =$ temperature.
- Heat flows from high to low.
- The Laplacian Δu measures the "average concavity" in all directions.

Heat Equation in Polar Coordinates

In polar coordinates (r, θ), the Laplacian changes form. The heat equation becomes:

$$\frac{\partial u}{\partial t} = \Delta_{\text{polar}} u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta}$$

- Same physics, different coordinates.
- Extra terms come from geometry.
- Laplacian depends on how we measure space.

Note

The Laplacian in Polar coordinates comes from the pushforward of the Cartesian Laplacian under the coordinate transformation. The extra terms arise because the basis vectors in polar coordinates are not constant and depend on position, leading to additional contributions when applying the chain rule.

Boundary Conditions

To solve the heat equation on a bounded domain, we must specify what happens at the boundary ∂M.

Dirichlet Boundary Condition

Fixes the temperature at the boundary:

$$u(x,t) = C \quad \text{for } x \in \partial M.$$

Physical meaning: The boundary is held at a constant temperature (e.g., submerged in ice water).

Neumann Boundary Condition

Fixes the heat flux across the boundary:

$$\frac{\partial u}{\partial n}(x,t) = C \quad \text{for } x \in \partial M.$$

Physical meaning: The boundary is perfectly insulated; no heat enters or escapes.

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What is a Metric?

- A metric tells us how to measure **distance, angle, and area**.
- It encodes the geometry of the space.

Examples

Flat (Cartesian):

$$ds^2 = dx^2 + dy^2.$$

Polar:

$$ds^2 = dr^2 + r^2 d\theta^2.$$

- Even flat space looks different under different coordinates.
- The metric determines how the second derivatives (and thus the Laplacian) are computed.

Laplacian Depends on Geometry

- In general, we write the Laplacian as Δ_g.
- The metric g determines how we compute derivatives.
- Different metrics $\Rightarrow$ different Laplacians.
- Therefore: geometry directly affects heat flow.

$$\text{metric} \to \text{Laplacian} \to \text{heat equation.}$$

Conformal Metrics and Weight Functions

Definition

A metric g is *conformal* to the flat metric if there exists a strictly positive weight function $u(x, y)$ such that:

$$g = u(x, y)(dx^2 + dy^2).$$

- In 1D, this looks like $g = u(x)dx^2$.
- The metric is obtained by scaling the flat metric.
- Angles are preserved, but lengths are rescaled by $\sqrt{u}$.
- Large u stretches distances; small u shrinks them.
- Geometry is encoded entirely in the weight function u.

The Graph of u vs. The Surface Itself

It is crucial to distinguish between the function $u(x, y)$ and the geometry it creates.

The Graph of u:

- The set of points $(x, y, u(x, y))$ in $\mathbb{R}^3$.
- The graph of u can be used to produce an embedding of the 2d manifold in 2d.

The Surface defined by the metric $g = u(x, y)(dx^2 + dy^2)$:

- You do not need a 3D ambient space to measure distances here.
- Walking on the xy-plane with a "stretchy ruler" defined by $u(x, y)$ is fundamentally different from walking on the 3D graph of u.

Fubini–Study Metric

A classic example of a conformal metric on the sphere $S^2 \cong \mathbb{C} \cup \{\infty\}$ is the Fubini-Study metric:

$$g_{FS} = \frac{4}{(1+x^2+y^2)^2}(dx^2+dy^2).$$

Here, our conformal weight function is:

$$u(x,y) = \frac{4}{(1+x^2+y^2)^2}.$$

This maps the flat plane to the standard round sphere via stereographic projection.

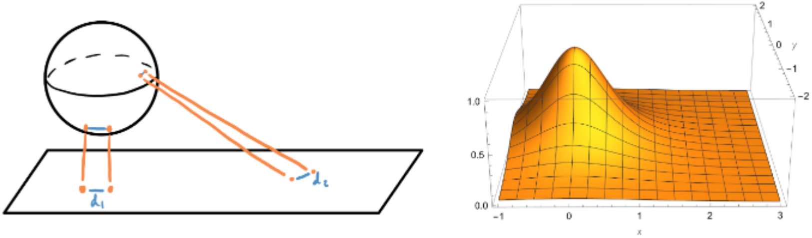

Figure: Fubini-Study metric visualized as a surface in $\mathbb{R}^3$. The geometry of the surface encodes the curvature of the sphere.

Curvature of the Fubini-Study Metric

For a conformal metric $g = u(x,y)(dx^2 + dy^2)$, the Gaussian curvature K is given by:

$$K = -\frac{1}{2u}\Delta(\log u).$$

Let's calculate K for Fubini-Study, where $u = 4(1+r^2)^{-2}$ and $r^2 = x^2 + y^2$:

$$\log u = \log 4 - 2\log(1+r^2)$$

$$\Delta(\log u) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(\log 4 - 2\log(1+r^2))$$

$$\partial_x \log(1+r^2) = \frac{2x}{1+r^2}$$

$$\begin{aligned}\partial_{xx}\log(1+r^2) &= \partial x\left(\frac{2x}{1+r^2}\right)\\ &= \frac{2(1+r^2)-4x^2}{(1+r^2)^2} = \frac{2-2x^2+2y^2}{(1+r^2)^2}\end{aligned}$$

$$\Delta\log(1+r^2) = \frac{2-2x^2+2y^2+2-2y^2+2x^2}{(1+r^2)^2} = \frac{4}{(1+r^2)^2}$$

$$\Delta(\log u) = -2\left(\frac{4}{(1+r^2)^2}\right) = -\frac{8}{(1+r^2)^2}.$$

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Curvature of the Fubini-Study Metric (Cont.)

Plugging the Laplacian back into our curvature formula:

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\begin{aligned}
K &= -\frac{1}{2u}\Delta(\log u) \\
&= -\frac{1}{2\left(\frac{4}{(1+r^2)^2}\right)}\left(-\frac{8}{(1+r^2)^2}\right) \\
&= \frac{(1+r^2)^2}{8}\cdot\frac{8}{(1+r^2)^2} \\
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The Fubini-Study metric describes a surface of **constant positive curvature** $K = 1$, which perfectly matches our geometric intuition of a standard sphere.

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Connections and the Geodesic Equation

How do things move straight in a curved space? We need a **connection** (∇) to tell us how vectors change as we move.
The metric determines the Levi-Civita connection, expressed via Christoffel symbols Γ_{ij}^k.

The Geodesic Equation

A path $\gamma(t)$ is a geodesic (a "straight line" in curved space) if its acceleration is zero with respect to the connection:

$$\nabla_{\dot\gamma}\dot\gamma = 0.$$

In local coordinates, this gives a system of 2nd-order ODEs:

$$\frac{d^2\gamma^k}{dt^2} + \sum_{i,j} \Gamma_{ij}^k \frac{d\gamma^i}{dt}\frac{d\gamma^j}{dt} = 0.$$

Given a conformal metric, we can compute the geodesic equation as:

$$\nabla_{\dot\gamma}\dot\gamma = \ddot\gamma + \left(\partial_z \log(u^2)\right)\dot\gamma^2 = 0.$$

The Ricci Flow Equation

Introduced by Richard Hamilton (1982), Ricci flow evolves the metric $g(t)$ to smooth out irregular curvature, governed by the Ricci curvature tensor Ric:

$$\frac{\partial}{\partial t} g_{ij} = -2\mathrm{Ric}_{ij} + \rho g_{ij}.$$

For 2D surfaces, $\mathrm{Ric}_{ij} = K g_{ij}$, where K is the Gaussian curvature. The scalar ρ dictates the type of flow:

- $\rho = 0$ **(Unnormalized Ricci Flow):** The space shrinks or expands depending on the sign of curvature. Positive curvature shrinks to a point.
- $\rho = \rho_t$ **(Normalized Ricci Flow):** Here, ρ_t is the time-dependent average curvature. This formulation preserves the total area of the surface as it evolves.
- $\rho =$ **constant (Yamabe Flow):** Evolving the metric by a constant scalar times the metric.

Note

Flowing the weight function $u(x, y, t)$ smooths out bumps in the actual embedded manifold. This happens with bubbles in 3D space, as they round out as much as possible given whatever boundary constraints are imposed.

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Ricci Flow on Surfaces

For our 2D conformal metric $g = u(x, y, t)(dx^2 + dy^2)$, the unnormalized Ricci flow equation simplifies drastically.
Since $\frac{\partial g}{\partial t} = -2Kg$, and $g = u(dx^2 + dy^2)$, the weight function evolves according to:

$$\frac{\partial u}{\partial t} = -2Ku.$$

Substitute $K = -\frac{1}{2u}\Delta(\log u)$, we get:

$$\frac{\partial u}{\partial t} = -2\left(-\frac{1}{2u}\Delta(\log u)\right) u = \Delta(\log u),$$

which is nice!